

# Practicum Sprint #6 - Final Delivery

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## 1 IMPLEMENTATION AND LINKS

### - External Links

- GitHub Repository: <https://github.gatech.edu/lalfaro8/medication-safety-assistant>
- Hosted Link: <https://medication-safety-assistant-production.up.railway.app/>
- Presentation Link:

### - Final Task Chart/List

| Task                             | Description                                                                                                                          | Needs/Risks                                                                                                             | Completion Status                                                                                                                    |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Research Medication Data Sources | Research public medication data sources including RxNorm/RxNav and OpenFDA to support medication lookup and get safety information.  | Need: Reliable public medication data<br>Risk: May not have full interaction data, requiring manual rule creation       | <b>Completed</b><br><br>Final system used drug label text and comparison logic since interaction data was limited.                   |
| Design UI Wireframes             | Create Figma wireframes for medication entry, profile setup, and warning displays                                                    | Risk: UI may need redesigns if requirements change                                                                      | <b>Completed</b><br><br>UI was expanded to dashboard interface while still following wireframes                                      |
| Set Up Application Architecture  | Set up the full stack application structure with React frontend, Flask backend API, SQLite database, and a medication comparison API | Need: Team agreement on stack and development workflow<br>Risk: Setup issues with React or backend/frontend integration | <b>Completed</b><br><br>Team was able to agree on stack early which includes React, Flask, RxNorm/RxNav, OpenFDA, and comparison API |

|                                   |                                                                                                                                                                                                     |                                                                                                                                                             |                                                                                                                                               |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Develop a User Authentication     | Implement user registration, login, logout, and protected user data access                                                                                                                          | Need: Secure access to specific user medications and profile data                                                                                           | <b><i>Completed</i></b><br><br>Final application uses JWT authentication for protected API routes                                             |
| Develop User Profile              | The app needs to be able to hold the user's medication information.                                                                                                                                 | Need: Deciding what data to store and what to display. Avoid information dumping, but need to ensure important information is still being given to the user | <b><i>Completed</i></b><br><br>User profile supports allergies, conditions, notes, medications, and favorite pharmacy selection.              |
| Develop Medication Profile        | This will be a combination of user input and editable information, as well as medication information that is NOT editable. Need to include basic drug information (how much, medication scheduling) | Risk: Input requirements may be more complex than expected                                                                                                  | <b><i>Completed</i></b><br><br>Medication cards display saved medication data, personal details, schedule information, and warning indicators |
| Integrate Medication Lookup API   | Connect backend to RxNorm/RxNav for medication lookup                                                                                                                                               | Need: Documentation and working API access<br>Risk: API limitations                                                                                         | <b><i>Completed</i></b><br><br>Users can search for medications and add/save selected results to their dashboard                              |
| Implement Medication Safety Rules | Implement backend logic for interaction checks and allergy conflicts                                                                                                                                | Risk: Curated medication rules could take longer to create and validate than expected                                                                       | <b><i>Completed</i></b><br><br>Application displays allergy and interaction warnings when possible concerns are found                         |

|                                      |                                                                                                       |                                                                                                                                                                                                                                  |                                                                                                                                         |
|--------------------------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Implement Time Conflict Analysis     | Implement logic for medications that should be spaced apart                                           | Need: Reliable timing rules for the selected medications<br>Risk: Limited public information available for some medications                                                                                                      | <i>Not Completed</i><br><br>Could not find public drug information interactions having to do with time                                  |
| Integrate Drug Label Comparison Data | Integrate OpenFDA label data through the medication comparison API to support medication label review | Need: Reliable safety medication label text<br><br>Risk: Label data may be incomplete, missing, or difficult to interpret                                                                                                        | <i>Completed</i><br><br>Final application uses the label text for warning explanations and medication comparison                        |
| Implement 'add medication'           | The user needs to be able to add to and edit their list of medications.                               | Risk: Cannot be implemented before gaining access to the medication database                                                                                                                                                     | <i>Completed</i><br><br>Users can search for medications and add them, which will trigger the safety checks                             |
| Implement Schedule Features          | Add medication schedule tracking with day and time fields for saved medications                       | Risk: Reliable clinical timing conflict rules were limited in public data                                                                                                                                                        | <i>Adjusted / Completed</i><br><br>Implementation of medication schedules was completed, but time conflict analysis was not implemented |
| Implement Warning System             | Create a frontend display for interaction alerts, allergy warnings, and timing conflicts              | Need: Make sure to reiterate that the app is NOT for medical advice, but still try to offer helpful warnings.<br>Risk: This is one of the last tasks because it needs the users' stored data and access to medication databases. | <i>Completed</i><br><br>Final app includes warning badges, warning summaries, expandable label details, and full warning label text     |

|                                      |                                                                                                  |                                                                                                             |                                                                                                                                                                                             |
|--------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implement Medication Comparison Tool | Add a medication comparison feature to review two medications without the need to save them      | Risk: Comparison depends on the available label data                                                        | <b>Completed</b><br><br>Users can enter two medications and review possible safety concerns                                                                                                 |
| Implement Pharmacy Search            | Add a pharmacy search and favorite pharmacy to save to a user's profile                          | Risk: Pharmacy search depends on Google Places API results                                                  | <b>Completed</b><br><br>Users can search for and save a favorite pharmacy to their profile.                                                                                                 |
| Create Synthetic Test Cases          | Create handcrafted patient scenarios with medication lists, allergies, and schedules for testing | Need: Representative test cases<br>Risk: test cases may not cover enough edge cases                         | <b>Completed</b><br><br>Testing focused on account creation, login, medication search, saved medications, allergy warnings, interaction warnings, schedules, and comparison                 |
| Integration Testing and Bug Fixing   | Test the full workflow from user input to backend analysis and display warnings                  | Risk: bugs could appear late in development                                                                 | <b>Completed</b><br><br>Fixed issues related to authentication using sessions and switched to JWT tokens, protected routes, deployment variables, API calls, and frontend warning behaviors |
| Deploy Application                   | Deploy the frontend, backend, and other needed services for public hosting                       | Risk: Deployment configuration issues could prevent the app from working just like the local implementation | <b>Completed</b><br><br>Application was deployed and tested through the hosted Railway link                                                                                                 |
| Prepare Final Delivery               | Prepare technical documentation, final report, and recorded presentation,                        | Risk: delays in previous tasks could result in less time to spend polishing                                 | <b>Completed</b><br><br>Final documentation and presentation materials were made                                                                                                            |

## 2 PROJECT SUMMARY AND DISCUSSION

### - Background and Need

Medication safety is a persistent challenge in modern healthcare, particularly as patients are increasingly prescribed multiple medications to manage chronic health conditions. Polypharmacy, the use of five or more medications, has been associated with higher risk of adverse drug events, medication non-adherence, and drug-to-drug interactions. According to the Centers for Disease Control and Prevention, data from 2017 to March 2020 indicate that 49.9% of adults in the United States used at least one prescription medication in the past 30 days, 24.7% used three or more, and 13.5% used five or more. As the number of medications increases, so does the potential for drug interactions, adverse effects, and the overall complexity of managing them safely.

One of the most concerning risks associated with polypharmacy, and the focus of our project, is the potential for harmful drug interactions. Drug interactions can lead to unexpected side effects, reduced therapeutic effectiveness, and in severe cases can require hospitalization. Research published in the *Journal of the American Medical Association* has shown that adverse drug events are responsible for approximately 10% of emergency department visits among older adults. Similarly, research supported by the National Institutes of Health suggests that a significant up to half of all adverse drug events are preventable and are often the result of inadequate medication monitoring, limited patient awareness, or gaps in communication.

Healthcare providers are responsible for ensuring that prescribed medications are safe to use in combination, however patients are often left to manage these medications independently. Medication instructions can be complex and difficult for non-expert users to interpret, increasing the likelihood of misunderstanding. Consequently, patients frequently rely on fragmented sources, such as prescription labels, online searches, or verbal instructions, in lieu of formal medical guidance.

This gap highlights the need for a user-friendly solution that empowers users to actively manage their medications and help inform them of potential risk. The goal of this project is to develop an intuitive application that allows users to track their medications, surface key information about each drug, and identify possible interactions in an understandable way. By leveraging publicly available healthcare data and aligning with interoperability standards such as HL7 International, this project aims to make complex

clinical information more accessible and support safer medication practices in everyday use.

## - **Solution**

DoseWise is a user-facing web application that helps individuals manage their medications and better understand potential risks. It provides a single interface where users can track medications, view key drug information, record allergies, and identify possible drug-to-drug interactions. The system is designed for non-clinical users, particularly those managing multiple prescriptions, and presents medication information in a more cohesive and understandable way.

Users can build and maintain a personalized list of medications by searching and adding drugs to their profile. Each medication is stored along with relevant information, and users can optionally include details such as dosage, timing, and personal notes to reflect their actual regimen.

Information is presented in a structured format that prioritizes readability, with expandable sections for more detailed content such as full medication labels. This allows users to access important details quickly without being overwhelmed by dense clinical text.

When a new medication is added, the system automatically evaluates it against the user's existing list. It performs pairwise interaction checks and cross-references recorded allergies to identify potential risks. If an issue is detected, the interface highlights the specific medications or allergens involved and provides a short, readable explanation. All information presented in the system is derived from official drug labeling and is intended for informational use only. DoseWise does not replace professional medical advice, but instead supports users in staying informed and having more productive conversations with healthcare providers.

The system consists of a frontend interface and a backend processing layer. The frontend handles user interaction, including search, medication management, and real-time display of interaction warnings. The backend manages data retrieval, normalization, and analysis, including interaction checks across the medication list and allergy conflict detection.

DoseWise integrates publicly available healthcare data sources to support these features. Medication search and normalization are handled through the National Library of Medicine RxNorm API, while drug safety and interaction-related information are

retrieved from the U.S. Food and Drug Administration OpenFDA API. These sources ensure that the data is standardized and grounded in official clinical documentation.

To align with modern healthcare data practices, the system incorporates elements of the HL7 International Fast Healthcare Interoperability Resources (FHIR) standard.

Medication and patient data are structured in a way that reflects common FHIR resource models, making the system easier to integrate with other healthcare technologies.

Overall, DoseWise brings medication tracking, interaction checking, and drug information into a single interface that is easier to navigate than existing fragmented sources. It is designed to help users better understand their medications and recognize potential risks in everyday use.

## - Discussion and Future Work

Overall, we believe this project was successful in its goal of supporting medication management outside of a clinical setting. The application is easy to use and does a good job of not only highlighting potential risks, but also explaining those risks to the user.

Future work may include improving how drug interactions are identified. Due to constraints in available resources (as students limited to publicly available data), the current system relies on drug label information to detect potential interactions. While this approach can surface useful signals, it is not a comprehensive method for identifying true drug–drug interactions. Incorporating a clinical database with reliable interaction data would significantly improve the system.

## - References

Centers for Disease Control and Prevention. (2023). *Therapeutic drug use*. National Center for Health Statistics.

<https://www.cdc.gov/nchs/fastats/drug-use-therapeutic.htm>

Budnitz, D. S., Shehab, N., Lovegrove, M. C., & Geller, A. I. (2011). Emergency hospitalizations for adverse drug events in older Americans. *Journal of the American Medical Association*, 305(19), 2002–2012. <https://doi.org/10.1001/jama.2011.672>

Tache, S. V., Sonnichsen, A., & Ashcroft, D. M. (2011). Prevalence of adverse drug events in ambulatory care: A systematic review. *Annals of Pharmacotherapy*, 45(7–8), 977–989. <https://doi.org/10.1345/aph.1P627>

### 3 TECHNICAL DOCUMENTATION

#### - User Manual

1. Access the Application
  - a. You can open the deployed DoseWise application through hosted link URL listed in Section 1.
2. Creating an account
  - a. You can click on “Need an account? Create one” to start creating an account
  - b. Enter a valid name, email address, and password
  - c. Click “Create Account”, after successful registration, the user will be automatically logged into the dashboard.
3. Logging In
  - a. Enter the email and password associated with the existing account.
  - b. Click “Log In”, user will be sent to the dashboard after a successful login.
4. Managing the User Profile
  - a. Click “View Profile” from the top of the dashboard.
  - b. Enter or update profile information such as name, age, allergies, and medical conditions.
  - c. Click “Save Profile” to save the changes.
  - d. When saving profile, DoseWise checks the user’s saved medications against the updated allergy checklist.
5. Searching for Medications
  - a. From the dashboard, use the “Search Medication” field to input a medication name and click “Search.”
  - b. Medication results will be displayed with available medication identifiers and metadata
6. Saving Medication
  - a. Click “Add” on a medication result you wish to save
  - b. The medication is then saved to the user’s dashboard

- c. DoseWise checks the newly added medication against the user's saved medications and allergy profile.
- d. If conflicts/concerns are found, warning messages will be displayed.

#### 7. Viewing Medication Warnings

- a. Saved medications may display interaction or allergy badges.
- b. Click a warning badge to view additional warning details.
- c. To view Label Highlights information of medication, click on "Show drug info."
- d. For interaction reports, use the "View label details" and "Show full warning label" options to read the label text

#### 8. Adding Medication Details

- a. On a saved medication card, click "Add personal details" or "Edit personal details."
- b. Enter dosage information and notes.
- c. Click "Save details."

#### 9. Adding a Medication Schedule

- a. Open the medication's personal details section.
- b. Select a day of the week and time.
- c. Click "Add Schedule."
- d. Existing schedules can be deleted with the "Remove" button.

#### 10. Deleting a Medication

- a. Click on "Delete" to remove medication from your dashboard.
- b. DoseWise checks your updated saved dashboard medication for conflicts/concerns.

#### 11. Comparing Medications

- a. Use the "Compare Medications" section.
- b. Enter two medication names.
- c. Click "Compare."
- d. DoseWise displays whether the medications should be reviewed and provides label comparison details when available

#### 12. Searching for a Favorite Pharmacy

- a. Open the profile page.
- b. Use the pharmacy search field to search for pharmacies by pharmacy name, city, or ZIP code.

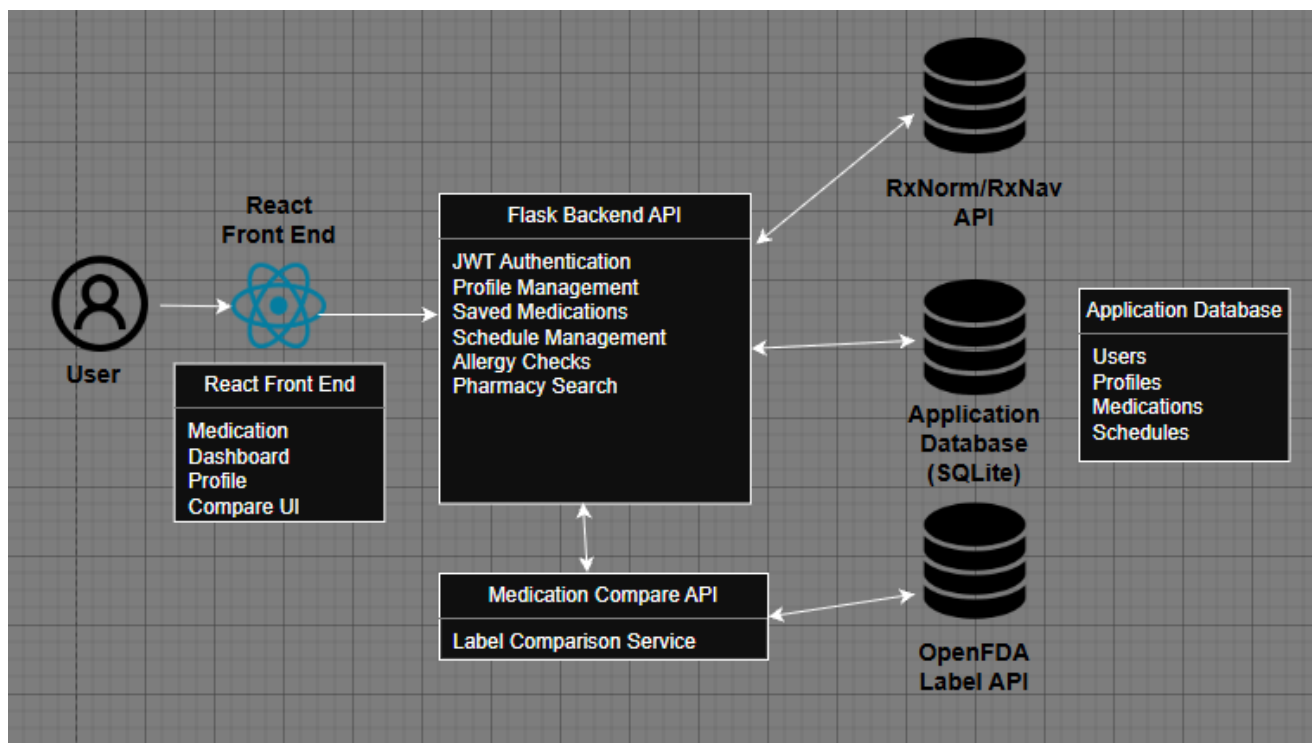
- c. Click "Set Favorite" on a result.
- d. To delete a favorite pharmacy, click "Remove Favorite Pharmacy" or set a new favorite pharmacy.
- e. Click "Save Profile" to save your changes.

### 13. Logging Out

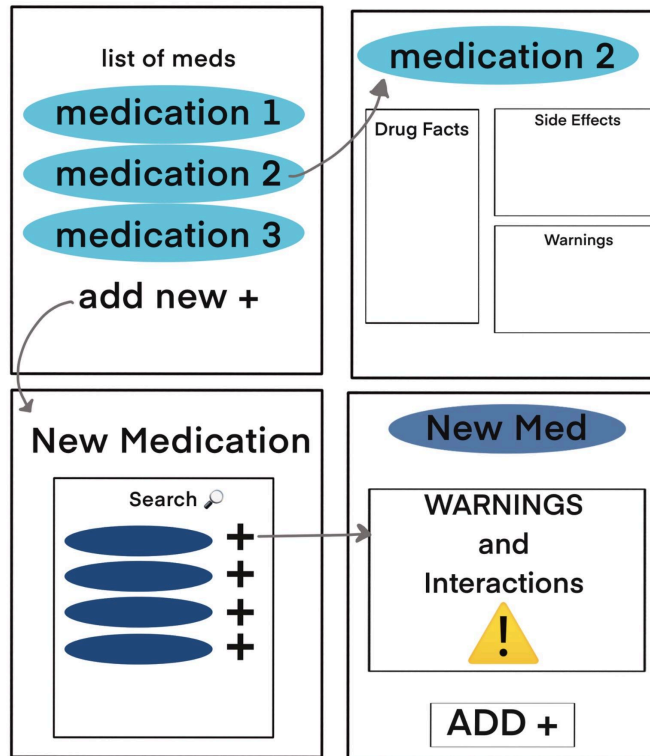
- a. Click "Log Out" at the top right of the dashboard

## - Diagrams and Screenshots

### - DoseWise system architecture



- Screen Mock up and Final Interface Screenshots



a. Searching for Medications

The screenshot shows the 'Search Medications' interface. The search bar contains the text 'Tylenol'. The results are as follows:

| Medication Name          | RxCUI   | Type | Action |
|--------------------------|---------|------|--------|
| Tylenol                  | 202433  | BN   | Add    |
| Tylenol Pill             | 1187315 | SBDG | Add    |
| Tylenol Oral Product     | 1187311 | SBDG | Add    |
| Tylenol Chewable Product | 1296526 | SBDG | Add    |

b. Medication Dashboard with drug facts, warnings, and interactions



# Medication Dashboard

Track medications, review label highlights, and stay on top of safety details.

Log Out

Medication check complete

×

Checked against 2 saved medications.

1 medication pairing should be reviewed:

- **ibuprofen** — ibuprofen appears in the drug interactions section of warfarin's label.

Hide label details

**Section:** Drug interactions

**Matched drug pair:** warfarin + ibuprofen

**Label excerpt:** 7 DRUG INTERACTIONS Concomitant use of drugs that increase bleeding risk, antibiotics, antifungals, botanical (herbal) products, and inhibitors and inducers of CYP2C9, 1A2, or 3A4. ( 7 ) Consult labeling of all concurren...

Show full warning label for warfarin

Show full warning label for ibuprofen

This label information comes from FDA drug labeling text returned through OpenFDA and compared for matching interaction-related language.

It should not be used as medical advice because label text can be incomplete, context-specific, and not tailored to a person's dose, history, or other medications.

WELCOME

bb

Add a favorite pharmacy and profile details for a more personalized experience.

View Profile

## Saved Medications

warfarin

Interaction

RxCUI: 11289

Type: IN

Add personal details

Show drug info

Delete

Benadryl

RxCUI: 203457

Type: BN

Add personal details

Show drug info

Delete

ibuprofen

Interaction

Interaction warning:

- Potential interaction with warfarin

RxCUI: 5640

Type: IN

Add personal details

Show drug info

Delete

c. Profile page with user information

[Back to Dashboard](#)

## Profile

**NAME**

Joe

**AGE**

56

**ALLERGIES**

☐ Milk (food)

☐ Egg (food)

☒ Fish (food)

☐ Crustacean shellfish (food)

☐ Tree nuts (food)

☐ Peanuts (food)

☐ Wheat (food)

☐ Soybeans (food)

☐ Sesame (food)

☒ Penicillin (drug)

**CONDITIONS**

Example: asthma, diabetes

**NOTES**

Add any personal notes

**Favorite Pharmacy**

**Capsule**

867 Peachtree St NE Suite 102, Atlanta, GA 30308, USA

(404) 800-0200

[Remove Favorite Pharmacy](#)

**Search for a Pharmacy**

Search by pharmacy name, city, or ZIP

[Search](#)

[Save Profile](#)

## d. Compare Two Medications

### Compare Medications

Run a side-by-side label comparison without changing your saved medication list.

Ibuprofen

Tylenol

Compare

#### Review recommended: No

- No direct interaction-related mentions were found in the fetched label sections.

This label information comes from FDA drug labeling text returned through OpenFDA and compared for matching interaction-related language.

It should not be used as medical advice because label text can be incomplete, context-specific, and not tailored to a person's dose, history, or other medications.

#### Hide full warning label for Ibuprofen

##### Warnings:

Warnings Allergy alert: Ibuprofen may cause a severe allergic reaction, especially in people allergic to aspirin. Symptoms may include: rash facial swelling asthma (wheezing) hives skin reddening shock blisters If an allergic reaction occurs, stop use and seek medical help right away. Stomach bleeding warning: This product contains an NSAID, which may cause severe stomach bleeding. The chance is higher if you: take more or for a longer time than directed take a blood thinning (anticoagulant) or steroid drug have had stomach ulcers or bleeding problems have 3 or more alcoholic drinks every day while using this product are age 60 or older take other drugs containing prescription or nonprescription NSAIDs [aspirin, ibuprofen, naproxen, or others] Heart attack and stroke warning: NSAIDs, except aspirin, increase the risk of heart attack, heart failure, and stroke. These can be fatal. The risk is higher if you use more than directed or for longer than directed. Do not use if you have ever had an allergic reaction to any other pain reliever/fever reducer right before or after heart surgery Ask a doctor before use if stomach bleeding warning applies to you you have a history of stomach problems, such as heartburn you have high blood pressure, heart disease, liver cirrhosis, kidney disease, asthma, or had a stroke you are taking a diuretic you have problems or serious side effects from taking pain relievers or fever reducers Ask a doctor or pharmacist before use if you are under a doctor's care for any serious condition taking aspirin for heart attack or stroke, because ibuprofen may decrease this benefit of aspirin taking any other drug When using this product take with food or milk if stomach upset occurs Stop use and ask a doctor if you experience any of the following signs of stomach bleeding: feel faint have bloody or black stools vomit blood have stomach pain that does not get better you have symptoms of heart problems or stroke: chest pain slurred speech trouble breathing leg swelling weakness in one part or side of body pain gets worse or lasts more than 10 days fever gets worse or lasts more than 3 days redness or swelling is present in the painful area any new symptoms appear If pregnant or breast-feeding, ask a health professional before use. It is especially important not to use ibuprofen at 20 weeks or later in pregnancy unless definitely directed to do so by a doctor because it may cause problems in the unborn child or complications during delivery. Keep out of reach of children. In case of overdose, get medical help or contact a Poison Control Center right away.

#### Hide full warning label for Tylenol

##### Warnings:

Warnings Liver warning This product contains acetaminophen. Severe liver damage may occur if you take more than 4,000 mg of acetaminophen in 24 hours with other drugs containing acetaminophen 3 or more alcoholic drinks every day while using this product Allergy alert: acetaminophen may cause severe skin reactions. Symptoms may include: skin reddening blisters rash If a skin reaction occurs, stop use and seek medical help right away. Do not use with any other drug containing acetaminophen (prescription or nonprescription). If you are not sure whether a drug contains acetaminophen, ask a doctor or pharmacist. If you are allergic to acetaminophen or any of the inactive ingredients in this product Ask a doctor before use if you have liver disease Ask a doctor or pharmacist before use if you are taking the blood thinning drug warfarin Stop use and ask a doctor if pain gets worse or lasts more than 10 days fever gets worse or lasts more than 3 days new symptoms occur redness or swelling is present These could be signs of a serious condition. If pregnant or breast-feeding, ask a health professional before use. Keep out of reach of children. Overdose warning In case of overdose, get medical help or contact a Poison Control Center right away. (1-800-222-1222) Quick medical attention is critical for adults as well as for children even if you do not notice any signs or symptoms.